Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/26/2022
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	Foam Optics and Mechanics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	djdurian@physics.upenn.edu	**Fax:**	FY 215-898-2010
PI Organization Type:	UNIVERSITY	**Phone:**	215-898-8147
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	209 South 33rd Street		
PI Web Page:	http://www.physics.upenn.edu/		
City:	Philadelphia	**State:**	PA
Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	02/20/2019	**End Date:**	03/25/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 03/25/2023 per NSSC information. (Ed. 11/24/21) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 9/9/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0599		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Continuation of "FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams)," grant NNX14AM99G, and previously "The Melting of Aqueous Foams," grant numbers NNX07AP20G ; NNX07AK60G ; NNC04GB61G. The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies Earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into Advanced Life Support (ALS) technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ALS, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During (and just before) this period we published papers outlining advances in three areas pertaining to the main focus of the grant. [Ed. note: See also Bibliography section for complete citations] The first (which appeared just prior to the current period but is not yet in Task Book) concerns the time evolution of individual bubble radii in wet foams that evolve by coarsening - the diffusion of gas from small high-pressure bubbles into large low-pressure bubbles. In particular, we present high-precision data for the time evolution of bubble area A(t) and circularity shape parameter C(t) for several bubbles in a quasi-two-dimensional foams consisting of bubbles squashed between parallel plates. In order to fully compare with earlier predictions, foam wetness is systematically varied by controlling the height of the sample above a liquid reservoir which in turn controls the radius r of the inflation of the Plateau borders. For very dry foams, where the borders are very small, classic von Neumann behavior is observed where a bubble’s growth rate depends only on its number n of sides. For wet foams, the inflated borders impede gas exchange and cause deviations from von Neumann’s law that are found to be in accord with the generalized coarsening equation. In particular, the overall growth rate varies linearly with the film height, which decrease as surface Plateau borders inflate. More interestingly, the deviation from dA/dt is proportional to (n - 6) von Neumann behavior grows in proportion to nCr/ Sqrt[A]. This is highlighted definitively by data for six-sided bubbles, which are forbidden to grow or shrink except for the existence of this term. It is tested quantitatively by variation of all four relevant quantities: n, C, r, and A. The second concerns a novel measurement of long-ranged structure in foams, believed to be important for underlying the flow rheology of bulk samples. In particular, we investigated the local and long-range structure of several space-filling cellular patterns: bubbles in a quasi- two-dimensional foam, and Voronoi constructions made around points that are uncorrelated (Poisson patterns), low discrepancy (Halton patterns), and displaced from a lattice by Gaussian noise (Einstein patterns). We study local structure with distributions of quantities including cell areas and side numbers. The former is the widest for the bubbles making foams the most locally disordered, while the latter show no major differences between the cellular patterns. To study long-range structure, we begin by representing the cellular systems as patterns of points, both unweighted and weighted by cell area. For this, foams are represented by their bubble centroids and the Voronoi constructions are represented by the centroids as well as the points from which they are created. Long-range structure is then quantified in two ways: by the spectral density, and by a real-space analog where the variance of density fluctuations for a set of measuring windows of diameter D is made more intuitive by conversion to the distance h(D) from the window boundary where these fluctuations effectively occur. The unweighted bubble centroids have h(D) that collapses for the different ages of the foam with random Poissonian fluctuations at long distances. The area-weighted bubble centroids and area-weighted Voronoi points all have constant h(D) = he for large D; the bubble centroids have the smallest value he = 0.084Sqrt [< a >], meaning they are the most uniform. Area-weighted Voronoi centroids exhibit collapse of h(D) to the same constant he = 0.084Sqrt [< a >] as for the bubble centroids. A similar analysis is performed on the edges of the cells and the spectra of h(D) for the foam edges

	show h(D) ~ D1-e where e = 0.30 ± 0.15. The third is a full scientific description of the Soft Matter Dynamics (FOAM-C) experimental container, which is now aboard the International Space Station (ISS). In this paper we describe an experiment container with light scattering and imaging diagnostics for experiments on soft matter aboard the ISS. The suite of measurement capabilities can be used to study different materials in exchangeable sample cell units. The currently available sample cell units and future possibilities for foams, granular media, and emulsions are presented in addition to an overview of the design and the diagnostics of the experiment container. First results from measurements performed on ground and during the commissioning aboard the ISS highlight the capabilities of the experiment container to study the different materials.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Chieco AT, Durian DJ. "Coarsening Behavior and Coarsening Induced Rearrangements in Very Wet Quasi-2D Foams." APS March Meeting 2021, Virtual, March 15-19, 2021. Bulletin of the American Physical Society. 2021 Mar;66(1):Abstract: P05.00008. https://meetings.aps.org/Meeting/MAR21/Session/P05.8 , Mar-2021
Articles in Peer-reviewed Journals	Chieco AT, Durian DJ. "Experimentally testing a generalized coarsening model for individual bubbles in quasi-two-dimensional wet foams." Phys Rev E. 2021 Jan;103(1):012610. https://doi.org/10.1103/PhysRevE.103.012610 ; PMID: 33601566 , Jan-2021
Articles in Peer-reviewed Journals	Chieco AT, Durian DJ. "Quantifying the long-range structure of foams and other cellular patterns with hyperuniformity disorder length spectroscopy." Phys Rev E. 2021 Jun;103(6-1):062609. https://doi.org/10.1103/PhysRevE.103.062609 ; PMID: 34271712 , Jun-2021
Articles in Peer-reviewed Journals	Born P, Braibanti M, Cristofolini L, Cohen-Addad, Durian DJ, Egelhaaf SU, Escobedo-Sánchez MA, Höhler R, Karapantsios TD, Langevin D, Liggieri L, Pasquet M, Rio M, Salonen A, Schröter M, Sperl M, Sütterlin R, Zuccolotto-Bernez AB. "Soft Matter Dynamics: a versatile platform to study dynamics in soft matter in microgravity." Revi Sci Instrum. 2021 Dec;92(12):124503. https://doi.org/10.1063/5.0062946 , Dec-2021